

**Exact Brillouin-Zone Moment Formulas for the Lattice
Bisognano–Wichmann
Normalization of Three-Dimensional Staggered Fermions**

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Abstract

The ground-state entanglement Hamiltonian (EH) of free staggered lattice fermions in $3 + 1$ dimensions, for a half-space cut, is known numerically to take the Bisognano–Wichmann (BW) linear form with a direction-dependent normalization: the nearest-neighbour bond coefficients grow as $\pi\xi/\lambda$ with $\lambda_x \neq \lambda_\perp$, a lattice (regulator) property rather than a physical temperature. We close the analytic problem of this normalization. First, an exact block reduction maps the half-space problem, transverse momentum by transverse momentum, onto semi-infinite one-dimensional staggered chains with mass $\mu(k_y, k_z) = \sqrt{\cos^2 k_y + \cos^2 k_z}$, verified by an operator-level sum rule to 4.5×10^{-7} . Second, the EH of the semi-infinite staggered chain is exactly known [Eisler, J. Stat. Mech. (2025) 013101]: $H_{\text{EH}} = 4\kappa K(\kappa') T$ with $\kappa = 1/\sqrt{1 + \mu^2}$. Translating to bond-gradient normalization gives a single prefactor function for both the hopping and the mass channel, $g(\mu) = g_m(\mu) = (2/\pi)\kappa K(\kappa') = 1/\text{AGM}(1, \sqrt{1 + \mu^2})$. The direction-dependent normalizations are then exact Brillouin-zone moments, $1/\lambda_x = \langle g(\mu) \rangle_{\text{BZ}}$ and $1/\lambda_\perp = \langle 2 \cos^2 k_y \cdot g(\mu) \rangle_{\text{BZ}}$, giving $\lambda_x = 1.185467287349258\dots$ and $\lambda_\perp = 1.229428764341310\dots$, in agreement with direct 3D lattice measurements to 6×10^{-7} and 1×10^{-6} . Third, we prove the sign of the anisotropy: since g is strictly decreasing and $\mathbb{E}[\cos^2 k_y] = 1/2$, the difference $r_\perp - r_x = 2 \text{Cov}(\cos^2 k_y, g)$ is negative by the Chebyshev correlation inequality, so $\lambda_\perp > \lambda_x$ is forced by the geometry of the channel weights; interval arithmetic with outward rounding certifies $\lambda_\perp - \lambda_x \geq 0.04391$. We delimit the claims explicitly: the 1D closed form is prior work; the contributions here are the exact 3D reduction, the BZ moment formulas with certified enclosures, the anisotropy theorem, and a machine-validated boundary between the two. The normalization λ is not an entanglement temperature (in $1 + 1\text{D}$ the same estimator gives $\lambda = 1$ from the first lattice layer) and its area-entropy analogue is not a Newton constant.

I. INTRODUCTION

For a relativistic QFT and a half-space (Rindler wedge), the Bisognano–Wichmann theorem [1] states that the modular (entanglement) Hamiltonian is the boost generator, $H = 2\pi \int x T_{00} dx$: a linear deformation of the physical Hamiltonian. On a lattice, the discretized version of this form is known to describe integrable chains in their gapped phase through the corner transfer matrix (CTM) correspondence [4, 5], and, for free-fermion chains,

exactly solvable commuting-operator structures exist for several geometries [6]. For *gapped* chains the BW form holds up to a *nonuniversal, dispersion-dependent prefactor* [2].

In a numerical program on 3+1D staggered fermions (quantum relational network repository, v22–v25 [8]), the half-space EH was computed directly from lattice correlation matrices with double-double precision and clamp-calibrated modular kernels. Three facts emerged: (i) the BW linear *form* survives to five digits near the cut; (ii) the normalization is direction dependent, $\lambda_x = 1.185468$ vs $\lambda_\perp = 1.229430$ (transverse), stable across scaling windows; (iii) the same estimator applied to two 1 + 1D discretizations (XX and Wilson) gives $\lambda = 1.0000\text{--}1.0002$ from the first layer, identifying λ as a property of the 3 + 1D staggered regulator, not a continuum temperature. An earlier numerological reading $\lambda = 32/27$ was rejected by direct large- N computation (deviation $187\times$ the convergence residual), and the associated “lattice entanglement temperature” and “bare Newton constant” interpretations were retired. This paper reports the analytic closure of what remains: the exact origin of $\lambda_x, \lambda_\perp$.

II. EXACT REDUCTION TO ONE-DIMENSIONAL STAGGERED CHAINS

The Hamiltonian is the standard staggered discretization on an L^3 lattice (open in x , periodic in y, z ; hopping $\pm 1/2$; η -phases $\eta_y = (-1)^x$, $\eta_z = (-1)^{x+y}$). Because the transverse part is x -independent, a transverse Fourier–taste rotation block-diagonalizes the problem: for each transverse momentum pair (k_y, k_z) in the halved Brillouin zone, the two taste bands reduce exactly to two decoupled 1D staggered chains with masses $\pm\mu$,

$$\mu(k_y, k_z) = \sqrt{\cos^2 k_y + \cos^2 k_z}, \quad H_{\text{chain}} = \sum_x \frac{1}{2} (c_x^\dagger c_{x+1} + \text{h.c.}) + \mu \sum_x (-1)^x c_x^\dagger c_x, \quad (1)$$

and charge conjugation $c \rightarrow (-1)^x c^\dagger$ maps $K(-\mu) = -\Sigma K(\mu) \Sigma$ ($\Sigma = \text{diag}(-1)^x$). The half-space EH bond profile is therefore an exact BZ average of chain EH profiles:

$$A_x(\xi) = \langle K_\mu^{\text{1D}}(\xi, \xi+1) \rangle_{\text{BZ}} \text{ (uniform)}, \quad K_y(\xi) = \langle (2 \cos^2 k_y / \mu) K_\mu^{\text{1D}}(\xi, \xi) \rangle_{\text{BZ}} \text{ (mass channel)}. \quad (2)$$

This reduction is verified as an operator-level **sum rule** at $N = 32$: the direct 3D half-space scan equals the 1D chain sum with $\max |\Delta| = 4.5 \times 10^{-7} / 1.1 \times 10^{-7} / 1.2 \times 10^{-7}$ ($A_x/K_y/B_z$) inside the certified window $\xi \leq 5$ (the floor is the clamp-boundary resolution of the double-double kernel, not the algebra) [claim QRN-GRAV-030 [8]].

Defining the normalizations by $K_{\text{NN}} = g(\mu) \pi \xi + b$ and $(-1)^x K_{xx} = g_m(\mu) 2\pi \xi_s \mu + b_m$ ($\xi_s = \xi - 1/2$ the half-site coordinate), the 3D normalizations become

$$\boxed{\frac{1}{\lambda_x} = \langle g(\mu) \rangle_{\text{BZ}}, \quad \frac{1}{\lambda_\perp} = \langle 2 \cos^2 k_y \cdot g_m(\mu) \rangle_{\text{BZ}}.} \quad (3)$$

III. THE CHAIN PREFACTOR IS EXACTLY KNOWN

The EH of the semi-infinite staggered chain was solved exactly by Eisler [2] [eqs. (64), (67), (72) there]: the correlation matrix commutes with the tridiagonal matrix T ($t_m = -m/2$, $d_m = (-1)^m \mu(m - \frac{1}{2})$) and

$$H_{\text{EH}} = 4\kappa K(\kappa') T, \quad \kappa = \frac{1}{\sqrt{1+\mu^2}}, \quad \kappa' = \frac{|\mu|}{\sqrt{1+\mu^2}}, \quad (4)$$

K being the complete elliptic integral of the first kind. Our chain differs by the local unitary $c_n \rightarrow (-1)^n c_n$ (hopping sign), which leaves the staggered term and the bond-gradient magnitudes invariant. Matching normalizations gives a single function for both channels:

$$\boxed{g(\mu) = g_m(\mu) = \frac{2}{\pi} \kappa K(\kappa') = \frac{2}{\pi} \frac{K(\kappa')}{\sqrt{1+\mu^2}} = \frac{1}{\text{AGM}(1, \sqrt{1+\mu^2})} = \frac{2}{\pi} \int_0^{\pi/2} \frac{d\theta}{\sqrt{1+\mu^2 \cos^2 \theta}}.} \quad (5)$$

The equality $g_m \equiv g$ — observed numerically at the 10^{-5} level before the identification — is thus a theorem: both channels carry the same nonuniversal prefactor. Notably $g(1) = 1/\text{AGM}(1, \sqrt{2})$ is Gauss's constant. Small- μ expansion: $g = 1 - \mu^2/4 + 9\mu^4/64 - 25\mu^6/256 + \dots$ (an analytic series in μ^2 ; earlier conjectured $\mu^2 \ln \mu$ terms do not exist). Large- μ : $g = (2/\pi|\mu|) \ln(4|\mu|) (1 + O(1/\mu^2))$. A pre-registered screening had rejected the spacing-type candidate $\pi/(2K(\kappa'))$ (drift 0.16% $\rightarrow$ 3.8% over $\mu \in [0.5, \sqrt{2}]$); the exact solution explains why: the EH level spacing factorizes as [prefactor $4\kappa K(\kappa')$] $\times$ [T -spacing $\pi/(2\kappa K(\kappa))$], and the screening candidate was the spacing function, not the operator prefactor. Verification battery (21 PASS): three-representation agreement to 3.4×10^{-14} , Gauss-constant anchor to 1 ulp, match to the lattice extraction $\max |\hat{g}/g - 1| = 1.18 \times 10^{-5}$ over the pre-registered window $\mu \geq 0.5$ (mass channel: 4.4×10^{-7}), and a mutation layer (five deliberate errors, all detected) [QRN-C0-007, QRN-GRAV-031 [8]].

IV. BRILLOUIN-ZONE MOMENTS: CERTIFIED VALUES

With g known, $r_x = 1/\lambda_x$ and $r_\perp = 1/\lambda_\perp$ are parameter-free numbers. Three implementations sharing no discretization or interpolation: **A** (no AGM): 3D torus midpoint rule of the single integrand $[1 + (\cos^2 k_y + \cos^2 k_z) \cos^2 \theta]^{-1/2}$; **B**: 2D BZ midpoint rule with f64 AGM; **R**: interval Riemann sum with outward rounding ($n = 2^{15}$ cells on $[0, \pi/2]^2$; interval π ; cosine via remainder-padded Taylor series — no libm trust; interval AGM; hierarchical summation; exact power-of-two division). A and B agree to 3.9×10^{-16} and both lie inside the rigorous enclosures of R:

	quantity floating (A = B)	certified enclosure (R)
r_x	0.843549215293854	[0.843545503984956, 0.843552926638375]
$r_\perp$	0.813385882130202	[0.813356957837832, 0.813414806840221]
λ_x	1.185467287349258	[1.185462071698428, 1.185472502995920]
$\lambda_\perp$	1.229428764341310	[1.229385046338885, 1.229472484821826]

The direct 3D lattice measurements ($\lambda_x = 1.185468$, $\lambda_\perp = 1.229430$, $N \leq 192$ block method) agree to 6.0×10^{-7} and 1.0×10^{-6} . The division of labour is explicit: *proof-grade* digits come from the interval enclosure (width $O(h)$: $7.4 \times 10^{-6}/5.8 \times 10^{-5}$), *reference* digits from two independent double-precision routes. The certificate JSON records formulas, domain, normalization, methods, enclosures, and SHA-256 hashes [QRN-GRAV-032, QRN-GRAV-033 [8]].

V. THE ANISOTROPY IS A THEOREM

Theorem 1. $r_\perp < r_x$, *i.e.* $\lambda_\perp > \lambda_x$.

Proof. Under the uniform BZ measure, $Y = \cos^2 k_y$ and $Z = \cos^2 k_z$ are independent with $\mathbb{E}[Y] = 1/2$, and $h(s) = g(\sqrt{s})$ is strictly decreasing ($g'(\mu) = -\mu \langle \cos^2 \theta (1 + \mu^2 \cos^2 \theta)^{-3/2} \rangle < 0$). Then $r_\perp - r_x = \langle 2Yh(Y+Z) \rangle - \langle h(Y+Z) \rangle = 2 \text{Cov}(Y, h(Y+Z))$. Conditioning on Z ($\mathbb{E}[Y|Z] = \mathbb{E}[Y]$ by independence), $\text{Cov}(Y, h(Y+Z)) = \mathbb{E}_Z[\text{Cov}(Y, h(Y+Z) | Z)]$, and for each z the function $h(Y+z)$ is strictly decreasing in Y , so each conditional covariance is negative by the Chebyshev correlation inequality. $\square$

Interval arithmetic seals the inequality numerically: $r_x - r_\perp \geq 0.0301307 > 0$, hence $\lambda_\perp - \lambda_x \geq 0.04391$ (certified). Physically: the transverse channel weights $2 \cos^2 k_y$ are largest exactly where the effective mass μ is largest and the prefactor g smallest — a negative covariance between channel weight and mass-dependent prefactor. The direction of the anisotropy is thus geometric necessity, not a Lorentz-breaking surprise.

VI. FINITE-SIZE BEHAVIOUR AND INSTRUMENT FLOOR

Redefining the finite-chain numerics as a *convergence test toward the exact g* (rather than a discovery tool) separates physics from instrument. Measured at $\mu \in \{0.5, 1, \sqrt{2}\}$, $N = 64 \dots 512$: the residual $\hat{g}_N/g - 1$ is **N -independent** (fitted exponent $p \approx 0$), of size $\leq 8.1 \times 10^{-6}$ — an instrument floor set by the fit window ($\xi^* = 5$ –6, the κ -budget edge of the clamped double-double kernel), not a finite-size effect; the mass channel reads the exact value to 5.3×10^{-7} . The gapped/critical crossover is organized by the single variable $x = N \operatorname{arsinh}(\mu)$ ($\xi_{\text{corr}} = 1/\operatorname{arsinh} \mu$): for $x \geq 25$ the extraction is converged ($|d| \leq 6 \times 10^{-6}$ observed), while for $x \leq 2$ the chain is effectively massless and the extraction reads the massless value — quantifying why small- μ points must be excluded from prefactor fits at fixed N . Gates are fail-closed: an insufficient trust window returns “no measurement” rather than a silently extrapolated fit, and a deliberately degraded kernel (coarse clamp pair) is verified to trigger exclusion [QRN-GRAV-034 [8]].

VII. SCOPE, PRIOR ART, AND NEGATIVE STATEMENTS

Prior art. The 1D closed form $g(\mu)$ — the exact EH of the semi-infinite staggered chain — is Eisler’s result [2], not ours. We claim: (i) the exact 3D→1D reduction with its operator-level sum rule; (ii) the BZ moment formulas $1/\lambda_x = \langle g \rangle$, $1/\lambda_\perp = \langle 2 \cos^2 k_y g \rangle$ and their certified evaluations; (iii) the anisotropy theorem; (iv) the identification $g_m \equiv g$ as the two-channel consequence of the single prefactor.

Negative statements (established in the same program, kept as constraints). λ is a regulator normalization: direction-, discretization- and operator-normalization-dependent. It is not an entanglement temperature (the same estimator yields $\lambda = 1$ in two 1 + 1D discretizations from the first layer [8]); $\lambda = 32/27$ is excluded ($N = 96$ direct reading, deviation $187 \times$

the residual); the associated area-coefficient combination $G_{\text{entropy}}^{\text{bare}}(a; \text{regulator, direction scheme})$ varies by 3.7% between direction schemes and is not a Newton constant. Nothing in this paper depends on, or supports, an induced-gravity interpretation.

Verification taxonomy. Of the 42 PASS gates behind this paper: mathematical identities (three-representation agreement, sum rule), known-value benchmarks (Gauss constant, Eisler prefactor), internal consistency (independent implementations, interval containment), negative controls (mutation layers, degraded-kernel exclusion, weight/measure mutants), and convergence/floor measurements. **None are frozen novel predictions** — this paper closes a normalization analytically; it does not predict new data.

VIII. REPRODUCIBILITY

All computations run from a dependency-free Rust repository (fixed seeds, built-in PASS/FAIL against exact values, machine-validated claim ledger with explicit assumptions and falsifiers; interval arithmetic implemented from scratch with outward rounding and libm-independent $\pi/\cos$). The frozen artifact set and its SHA-256 manifest are produced by `v252_manifest`; the certificate is `results/v252_bz_certificate.json` [8].

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